

Sustainable Livestock Management Best Practices: An Overview of Piggery, Goat, and Rabbit Farming

Introduction

Sustainable livestock management encompasses a set of practices aimed at optimizing productivity while minimizing environmental impact and ensuring animal welfare. This comprehensive guide focuses on sustainable practices in piggery, goat, and rabbit farming, efficient waste management, breeding techniques for high productivity, and ethical farming standards.

Topics Covered

Sustainable Practices in Piggery, Goat, and Rabbit Farming

Piggery Farming

Spacious and Well-Ventilated Housing: Providing pigs with ample space and proper ventilation is crucial. Adequate ventilation prevents respiratory issues and reduces stress among the pigs. Housing should allow for free movement to ensure comfort and well-being. Ventilation can be achieved through natural airflow or mechanical systems, depending on climate and design (Danish Pig Academy, n.d.) and (Farm and Chill, n.d.).

Bedding Materials: Using bedding materials like straw or sawdust enhances pig comfort by providing a softer surface and absorbing waste, maintaining a cleaner environment. Consider factors such as cost, availability, and ease of management when choosing bedding (Danish Pig Academy, *ibid*) and (Farm and Chill, *ibid*).

Location Selection: Choose a location with good drainage to prevent waterlogging and reduce disease spread. Ensure the location is accessible for transporting feed and pigs and is situated away from residential areas to minimize odor and noise issues (*ibid*).

Temperature Control: Maintaining a comfortable temperature is vital. Pigs are sensitive to extreme temperatures, so housing should provide adequate insulation and protection from harsh weather. In hot climates, ensure ample shade and

airflow; in cold climates, provide draft-free shelters with sufficient insulation (ibid).

Flooring and Drainage: Flooring should be durable and non-slip to ensure pig safety. A well-designed drainage system keeps housing clean and dry, preventing health problems related to poor hygiene (ibid).

Feeding

Balanced Diet: Implement a balanced diet using locally available ingredients to reduce feed costs and support local agriculture. Incorporating crop residues and food waste into pig diets promotes circular economy practices, reducing waste and enhancing sustainability (Danish Pig Academy, ibid) (Farm and Chill, ibid).

Local Ingredients: Using locally sourced feed ingredients not only reduces costs but also supports local farmers. This practice promotes a symbiotic relationship between crop and livestock farming, where crop residues can be recycled into feed (ibid).

Circular Economy Practices: Incorporating food waste into pig diets is a sustainable practice that reduces overall waste and creates a circular economy. This method is beneficial for both environmental sustainability and cost reduction (Danish Pig Academy) and (Farm and Chill, ibid).

Health Management

Vaccination and Deworming: Following a strict vaccination and deworming schedule is essential to prevent diseases. Regular vaccinations protect pigs from common illnesses, while deworming ensures they remain free of internal parasites (ibid).

Health Checks and Biosecurity: Regular health checks help in early detection and management of health issues. Implementing biosecurity measures, such as footbaths, disinfection stations, and restricted access to the farm, limits the spread of diseases (ibid).

Biosecurity Protocols: Implementing strict biosecurity protocols is crucial. These include limiting farm access, disinfecting equipment and vehicles entering the farm, and isolating new animals before introducing them to the herd (ibid).

It is important for pig farmers to adhere to these best practices in housing, environment, feeding, and health management, in order to create a comfortable,

healthy living space for pigs. It also provides for a sustainable and productive piggery operation, promoting animal welfare and farm profitability.

Goat Farming

Housing and Pasture Management

Rotational Grazing: Implementing rotational grazing helps maintain pasture health by preventing overgrazing and promoting the regrowth of forage plants. This involves dividing the pasture into smaller sections and rotating goats between these sections to ensure they always have access to fresh forage while allowing grazed sections to recover (ibid).

Constructing Shelters: Shelters should protect goats from extreme weather conditions, such as excessive heat, cold, and rain. They should be spacious, well-ventilated, and allow goats to exhibit natural behaviors like climbing and playing. Use durable materials like wood and metal for construction.

Location Selection: Choose a location with good drainage to prevent waterlogging and reduce disease spread. Ensure accessibility for transporting feed and goats, and situate away from residential areas to minimize odor and noise.

Temperature Control: Maintaining a comfortable temperature is vital. Shelters should provide adequate insulation and protection from harsh weather. Ensure shade and airflow in hot climates, and draft-free shelters with insulation in cold climates.

Flooring and Drainage: Flooring should be durable and non-slip to ensure goat safety. A well-designed drainage system keeps housing clean and dry, preventing health problems related to poor hygiene.

Nutrition

Balanced Diet: Goats require a diet rich in roughage, including hay, silage, and pasture plants, supplemented with minerals and vitamins. A balanced diet ensures healthy growth, reproduction, and milk production. Always provide clean, fresh water for them.

Browse Plants and Fodder Trees: Incorporating browse plants and fodder trees into the diet can reduce feed costs and provide additional nutrients. Plants like acacia, mulberry, and moringa are excellent sources of protein and other nutrients, enhancing goat health.

Health Management

Monitoring for Parasites and Diseases: Regular health checks are essential for monitoring parasites and diseases. Common goat diseases include foot rot, mastitis, and pneumonia, while internal parasites like worms can cause significant health issues. Implement preventative measures, such as deworming and vaccination.

Natural Remedies and Supplements: Using natural remedies and supplements can boost goat immunity and reduce dependency on antibiotics. Herbal supplements, essential oils, and natural tonics can improve overall health and resistance to diseases. Practices like proper hygiene, clean bedding, and stress reduction are also crucial for maintaining healthy goats.

It is important for goat farmers to follow these best practices in housing, pasture management, nutrition, and health management to ensure sustainable and productive goat farming operations. This will in turn promote animal welfare and farm profitability.

Rabbit Farming

Housing

Hutches: Use hutches that offer adequate space, ventilation, and protection from predators. Ideal hutches should be raised off the ground to prevent dampness and pest infestations. Construct hutches with durable materials such as wire mesh and wood to ensure longevity and security.

Age Group Separation: Implement systems to separate different age groups, such as kits, growers, and adults, to prevent overcrowding and reduce stress. Separation also helps manage feeding and health care more effectively, ensuring each group receives appropriate care.

Space Requirements: Ensure that each rabbit has enough space to move freely and exhibit natural behaviors. Overcrowded conditions can lead to stress and health issues, negatively impacting productivity and welfare.

Ventilation: Good ventilation in hutches helps maintain air quality and reduce humidity, preventing respiratory problems. Ensure that hutches are well-ventilated but not drafty to maintain a stable environment.

Feeding

Balanced Diet: Provide a diet rich in fiber from hay and leafy greens, supplemented with commercial rabbit pellets. Fiber is crucial for healthy digestion and preventing gastrointestinal issues. Include a variety of fresh vegetables to ensure a balanced intake of nutrients.

Clean Water: Ensure constant access to clean and fresh water. Regularly clean and refill water containers to prevent contamination and ensure that rabbits stay hydrated, which is essential for their overall health and digestion.

Feed Quality: Use high-quality commercial rabbit pellets that are specifically formulated to meet the nutritional needs of rabbits. Avoid feeds that contain excessive grains or seeds, which can lead to obesity and other health issues.

Health Management

Hygiene Practices: Maintain strict hygiene practices to prevent common diseases like coccidiosis. Regularly clean hutches and remove waste to minimize the risk of infections. Use disinfectants safe for rabbits to ensure a clean environment.

Vaccination and Health Checks: Follow a vaccination schedule and conduct regular health checks to ensure the herd's health. Monitor rabbits for signs of illness, such as changes in appetite, behavior, or physical condition, and consult a veterinarian when necessary.

Disease Prevention: Implement biosecurity measures to prevent disease outbreaks. Limit access to the rabbitry, use footbaths, and ensure new rabbits are quarantined before introducing them to the main herd.

In adhering to these best practices in housing, feeding, and health management, rabbit farmers can create a sustainable and productive farming operation that prioritizes welfare and productivity of their rabbits at the same time.

Efficient Waste Management and Recycling

Composting

Creating Organic Fertilizer:

- **Manure Composting:** By composting manure, farmers can produce organic fertilizer that can be used to enrich pastures or sold to local farmers. This process not only helps in managing waste but also adds value to the farm's output by creating a marketable product. Composting transforms animal waste into a nutrient-rich soil amendment that improves soil health and fertility (Danish Pig Academy, n.d.) (Farm and Chill, n.d.)
- **Pasture Enrichment:** Use the composted manure to enhance the quality of pastures, ensuring a sustainable cycle of nutrients within the farm. The compost improves soil structure, increases water retention, and provides essential nutrients for pasture plants (Farm and Chill, ibid).

Aerobic and Anaerobic Composting Techniques:

- **Aerobic Composting:** This method involves the breakdown of organic materials by microorganisms that require oxygen. It is typically faster and results in fewer odors compared to anaerobic composting. Aerobic composting requires regular turning of the compost pile to ensure that oxygen is available throughout the material. This process helps in producing high-quality compost efficiently (Danish Pig Academy, ibid) and (Farm and Chill, ibid).
- **Anaerobic Composting:** This method involves decomposition by microorganisms that do not require oxygen. While it can take longer and produce more odors, it is suitable for managing large volumes of waste with less frequent turning. Anaerobic composting can be conducted in enclosed systems to minimize odors and capture methane for energy production (ibid).
- **Implementing Composting Systems:**
- **Compost Bins and Piles:** Set up compost bins or piles that are properly aerated to facilitate aerobic composting. Ensure that the composting site is

easily accessible and adequately sized to handle the volume of waste produced by the farm.

- **Monitoring and Management:** Regularly monitor the temperature, moisture, and aeration of the compost piles to ensure optimal conditions for microbial activity. Adjust the composting process as needed to maintain efficiency and prevent issues such as odor or incomplete decomposition. Take a look at the following provided steps as a guide.

Step-by-Step Guide for Creating or Implementing Composting Systems

1. Site Selection

- **Materials Needed:** Measuring tape, stakes, marking paint

2. Building Compost Bins or Piles

- **Materials Needed:** Lumber or pallets (for bins), wire mesh, nails or screws, hammer or screwdriver

3. Gathering Compost Materials

- **Materials Needed:** Manure, straw or sawdust, green waste (grass clippings, vegetable scraps), brown waste (leaves, wood chips)

4. Layering the Compost

- **Materials Needed:** Shovel, pitchfork, water source

5. Aerating the Compost

- **Materials Needed:** Compost turner or pitchfork

6. Moisture Control

- **Materials Needed:** Water source, tarp or compost cover

7. Monitoring the Compost

- **Materials Needed:** Compost thermometer, moisture meter

8. Harvesting the Compost

- **Materials Needed:** Shovel, wheelbarrow, sieve (optional)

In implementing these composting techniques, farms can efficiently manage waste, reduce environmental impact, and produce valuable organic fertilizer that enhances soil health and productivity.

Biogas Production

Anaerobic Digesters:

- **Manure Conversion:** Utilize anaerobic digesters to convert manure into biogas, a renewable energy source. This process involves the microbial breakdown of organic matter in the absence of oxygen, resulting in the production of methane-rich biogas (Danish Pig Academy, *ibid*) .
- **Energy Use:** The biogas produced can be used for various energy needs on the farm, such as heating, electricity generation, and cooking, thereby reducing reliance on fossil fuels and lowering energy costs (*ibid*).

Implementing Biogas Systems

- **Digester Setup:** Install an anaerobic digester system that is appropriately sized for the volume of manure produced on the farm. Ensure that the system includes all necessary components, such as the digester tank, gas storage, and piping.
- **Feedstock Management:** Regularly feed the digester with manure and other suitable organic materials. Maintain a balanced input to ensure optimal biogas production ([ibid](#)).

Materials Needed for Biogas Production:

- **Anaerobic Digester:** Digester tank, gas storage tank, piping, biogas stove or generator
- **Feedstock:** Manure, organic waste
- **Monitoring Equipment:** Temperature gauge, pressure gauge, gas flow met

Step-by-Step Guide for Biogas Production

1. Site Selection

- **Materials Needed:** Measuring tape, stakes, marking paint

2. Digester Construction

- **Materials Needed:** Concrete or steel for digester tank, gas storage tank, piping, biogas stove or generator, construction tools (shovel, hammer, etc.)

3. Gathering Feedstock

- **Materials Needed:** Manure, organic waste (crop residues, kitchen waste)

4. Loading the Digester

- **Materials Needed:** Shovel, wheelbarrow, water source

5. Maintaining Optimal Conditions

- **Materials Needed:** Temperature gauge, pressure gauge, pH meter

6. Biogas Collection and Storage

- **Materials Needed:** Gas storage tank, piping, gas flow meter

7. Using the Biogas

- **Materials Needed:** Biogas stove or generator, connection hoses

8. Monitoring and Maintenance

- **Materials Needed:** Regular monitoring schedule, spare parts for digester, maintenance tools

Recycling Water

Systems Implementation:

- **Cleaning Processes:** Install water recycling systems to capture, filter, and reuse water used in cleaning barns, equipment, and animal housing. This can significantly reduce the farm's overall water consumption.

- **Cooling Processes:** Utilize closed-loop cooling systems where water used for cooling can be recirculated after being treated to remove contaminants and reduce temperature ([Danish Pig Academy, n.d.](#)) and ([Farm and Chill, n.d.](#))

Materials Needed for Recycling Water Systems:

- **Filtration System:** Filters, pumps, storage tanks
- **Treatment Equipment:** UV sterilizers, chlorinators, pH adjusters
- **Plumbing Supplies:** Pipes, fittings, valves

Step-by-Step Guide for Recycling Water

1. Assessment and Planning

- **Materials Needed:** Site survey tools, water usage data, planning software

Recommended Software:

- **AutoCAD:** Used for creating detailed plans and layouts of the water recycling system.
- **WaterGEMS:** A hydraulic modeling application for water distribution networks.
- **EPANET:** A software application used to model water distribution piping systems.
- **Step:** Conduct a thorough assessment of water usage in cleaning and cooling processes. Plan the system layout and requirements based on the farm's needs.

2. Install Water Recycling Systems for Cleaning Processes

- **Materials Needed:** Filters, pumps, storage tanks, pipes, fittings, valves
- **Step:** Install a water recycling system to capture water used in cleaning barns, equipment, and animal housing. Ensure the system includes effective filtration to remove contaminants.

3. Install Closed-Loop Cooling Systems

- **Materials Needed:** Cooling towers, heat exchangers, pumps, storage tanks, pipes, fittings, valves
- **Step:** Set up a closed-loop cooling system to recirculate water used for cooling. Include treatment equipment to maintain water quality.

4. Filtration System Setup

- **Materials Needed:** Filters (mechanical, biological, and chemical), pumps, storage tanks
- **Step:** Install filtration systems to clean captured water. Use mechanical filters to remove large particles, biological filters for organic matter, and chemical filters to remove dissolved substances.

5. Water Treatment

- **Materials Needed:** UV sterilizers, chlorinators, pH adjusters
- **Step:** Incorporate treatment equipment to ensure water is safe for reuse. Use UV sterilizers to kill pathogens, chlorinators to disinfect, and pH adjusters to maintain the correct pH balance.

6. Plumbing and Integration

- **Materials Needed:** Pipes, fittings, valves, connection tools
- **Step:** Integrate the filtration and treatment systems into the farm's existing plumbing. Ensure all connections are secure and leak-free.

7. Testing and Calibration

- **Materials Needed:** Water quality testing kits, calibration tools
- **Step:** Test the water quality at various points in the system to ensure it meets safety standards. Calibrate equipment as necessary to maintain optimal performance.

8. Monitoring and Maintenance

- **Materials Needed:** Monitoring equipment, maintenance tools, replacement parts

- **Step:** Regularly monitor the system's performance and conduct routine maintenance. Replace filters and other components as needed to keep the system running efficiently.

Waste Minimization

How to Design Feeding Systems to Minimize Feed Wastage and Ensure Efficient Nutrient Utilization

1. Precision Feeders:

- Use automated feeders that dispense precise amounts of feed to reduce wastage. This ensures that each animal receives the right quantity of feed without excess being spilled or left uneaten.
- **Example:** Automatic feed dispensers for pigs, which release feed based on the pigs' weight and growth stage, minimizing overfeeding and wastage.

2. Feed Bunk Management:

- Design feed bunks to reduce spillage and prevent animals from trampling the feed. Raised feed bunks or feeders with barriers can help keep feed clean and accessible.
- **Example:** Cattle feed bunks with sloped sides that direct feed back towards the animals, reducing wastage.

3. Ad Libitum vs. Scheduled Feeding:

- Evaluate the benefits of ad libitum (free access) feeding versus scheduled feeding times. Ad libitum feeding can reduce competition and stress among animals, but scheduled feeding can help control intake and minimize waste.
- **Example:** Implementing scheduled feeding for poultry to prevent overconsumption and feed spillage.

4. Feeding Frequency and Portions:

- Increase feeding frequency with smaller portions to ensure animals consume all the feed provided. This helps in preventing large amounts of feed from going stale or being wasted.
- **Example:** Splitting daily feed rations for goats into multiple smaller feedings throughout the day.

5. Feed Formulation and Quality:

- Use high-quality feed formulations that match the nutritional needs of different livestock at various growth stages. Properly balanced diets ensure efficient nutrient utilization, reducing the amount of uneaten or poorly digested feed.
- **Example:** Customizing feed formulas for dairy cattle based on their lactation cycle to maximize milk production and nutrient absorption.

6. Feed Storage and Handling:

- Implement proper storage practices to maintain feed quality and prevent spoilage. Use covered bins or silos to protect feed from moisture and pests.
- **Example:** Installing silos with airtight seals to keep pig feed dry and pest-free.

7. Behavioral Considerations:

- Consider the natural feeding behaviors of the livestock to design systems that accommodate their habits. For instance, pigs are rooters and need feeders that can withstand their rooting behavior without spilling feed.
- **Example:** Heavy-duty feeders for pigs that can handle their rooting behavior and prevent spillage.

Breeding Techniques for High Productivity

Breeding techniques are essential for enhancing productivity, ensuring the sustainability of livestock operations, and improving overall herd health. Here's a

detailed guide to effective breeding techniques for high productivity in livestock farming:

1. **Selective Breeding:**

- **Genetic Selection:**
- Use genetic selection to improve desirable traits such as growth rate, feed efficiency, and disease resistance. Select animals with superior genetic traits to become parents of the next generation.
- Implement performance recording and data analysis to inform breeding decisions. Regularly assess animals for their genetic potential and productivity, using tools like Estimated Breeding Values (EBVs) and genetic markers.
- **Example:** In dairy cattle, selecting bulls with high milk production traits can improve milk yield in the herd.

2. **Crossbreeding:**

- Combine different breeds to enhance hybrid vigor, also known as heterosis, which can lead to improved productivity and resilience.
- Crossbreeding can result in offspring with better growth rates, higher fertility, and greater disease resistance.
- **Example:** Crossbreeding beef cattle breeds like Angus and Hereford to produce calves with better growth rates and meat quality.

3. **Artificial Insemination (AI):**

- Use artificial insemination to introduce superior genetics into the herd without the need for live animals, reducing the risk of disease transmission.
- AI allows for the use of semen from genetically superior males that may not be physically present on the farm, ensuring a wider gene pool and improved genetic quality.
- **Example:** In swine production, AI can be used to improve litter size and growth rates by using semen from high-performing boars.

Additional Techniques:

4. Embryo Transfer:

- Use embryo transfer to multiply the offspring of genetically superior females. This technique involves collecting fertilized embryos from donor females and implanting them into surrogate mothers.
- **Example:** In cattle, embryo transfer can significantly increase the number of calves from top-quality cows.

5. Genomic Selection:

- Incorporate genomic selection to identify animals with the best genetic makeup for breeding. This involves analyzing DNA to predict the performance of breeding stock more accurately.
- **Example:** Using genomic data to select dairy cows with higher milk yield and better health traits.

Animal Welfare and Ethical Farming Standards

Implementing high standards of animal welfare and ethical farming practices is essential for ensuring the health and well-being of livestock. It also fosters consumer trust and supports sustainable farming. Here's a detailed guide on best practices for animal welfare and ethical farming:

1. Humane Treatment:

- **Adhere to Humane Handling Practices:**
 - Ensure all handling practices minimize stress and injury to animals. This includes gentle handling, appropriate transportation, and careful handling during medical procedures.
- **Provide Enrichment Activities:**
 - Offer enrichment activities that promote natural behaviors and mental well-being, such as providing objects for exploration, space for movement, and opportunities for social interactions.

2. Ethical Slaughter:

- **Follow Ethical Slaughter Practices:**

- Implement humane slaughter methods that minimize pain and distress, in compliance with regulatory standards. This includes proper stunning techniques and ensuring that animals are handled calmly before slaughter.

3. Certification:

- **Obtain Relevant Certifications:**

Acquire certifications such as Organic, Free-Range, or Animal Welfare Approved to demonstrate a commitment to high ethical standards. In Nigeria, relevant certifications include:

- **Nigeria Organic Agriculture Network (NOAN) Certification:** Ensures organic farming practices.
- **Global Animal Partnership (GAP) Certification:** Promotes animal welfare standards.
- **Animal Welfare Approved by A Greener World (AGW):** Recognizes high welfare practices.

4. Education and Training:

- **Provide Ongoing Education and Training:**

- Regularly train farm staff on best practices in animal welfare and sustainable farming. This includes workshops, certifications, and continuous learning opportunities to stay updated with the latest welfare standards and practices.

Conclusion

Adopting sustainable livestock management practices is essential for ensuring long-term productivity, environmental stewardship, and animal welfare. By focusing on efficient waste management, ethical breeding techniques, and maintaining high standards of animal care, farmers can achieve a sustainable and profitable livestock operation. Key practices include:

- **Efficient Waste Management:** Implementing composting and biogas production to recycle waste and generate renewable energy helps reduce environmental impact and promotes a circular economy.

- **Ethical Breeding Techniques:** Utilizing selective breeding, crossbreeding, and artificial insemination can improve productivity, disease resistance, and overall livestock health while reducing reliance on antibiotics.
- **High Standards of Animal Care:** Adhering to humane treatment, ethical slaughter practices, and obtaining relevant animal welfare certifications ensure the well-being of animals and meet consumer demand for ethically produced products.
- **Resource Conservation:** Employing water and feed efficiency strategies conserves resources, lowers costs, and enhances farm sustainability.
- **Education and Training:** Ongoing education and training for farm staff on sustainable and ethical practices ensure continuous improvement and adherence to best practices.

By integrating these sustainable practices, farmers will not only enhance their operations but will also contribute to global efforts in promoting environmental sustainability and animal welfare. This is the direction where modern agriculture should be heading to in solidarity with the fight against climate change.

References

Animal Welfare Approved by A Greener World. (n.d.). Standards. Retrieved from <https://agreenerworld.org/animal-welfare-approved/>

Biogas World. (n.d.). Biogas production. Retrieved from <https://biogasworld.com>

Cornell Waste Management Institute. (n.d.). On-farm composting of livestock manure. Retrieved from <https://cwmi.css.cornell.edu>

Danish Pig Academy. (n.d.). Housing technology in pig farming: Provide a comfortable and healthy living environment for your pigs. Retrieved from <https://www.danishpigacademy.com/>

Danish Pig Academy. (n.d.). Water Recycling in Piggery Farming. Retrieved from <https://www.danishpigacademy.com>

EPA. (n.d.). Water Recycling and Reuse. Retrieved from <https://www.epa.gov>

FAO. (n.d.). Water reuse. Retrieved from <https://www.fao.org>

Farm and Chill. (n.d.). Efficient Water Use in Livestock Farming. Retrieved from <https://www.farmandchill.com>

Farm and Chill. (n.d.). Farm pig housing: Ultimate guide for building a pig shelter. Retrieved from <https://www.farmandchill.com/>

Global Animal Partnership. (n.d.). Animal Welfare Certification Standards. Retrieved from <https://globalanimalpartnership.org/standards/>

Iowa State University Extension and Outreach. (2021). Feed Bunk Management and Efficiency. Retrieved from <https://www.extension.iastate.edu/news/feed-bunk-management-efficiency>

Merck Veterinary Manual. (n.d.). Rabbit management and nutrition. Retrieved from <https://www.merckvetmanual.com/>

National Research Council. (1998). Nutrient Requirements of Swine. 10th Edition. Washington, DC: The National Academies Press. <https://doi.org/10.17226/6016>

Nigeria Organic Agriculture Network. (n.d.). Certification. Retrieved from <https://www.noanigeria.net/certification/>

Penn State Extension. (n.d.). Anaerobic Digestion: Biogas Production and Odor Reduction from Manure. Retrieved from <https://extension.psu.edu>

Penn State Extension. (n.d.). Goat production: Best practices for housing, feeding, and health management. Retrieved from <https://extension.psu.edu/>

Penn State Extension. (n.d.). Raising small groups of pigs. Retrieved from <https://extension.psu.edu>

Purdue University Extension. (n.d.). Feeding and nutrition for rabbits. Retrieved from <https://extension.purdue.edu>

Renewable Energy Policy Network for the 21st Century (REN21). (n.d.). Biogas technology and management. Retrieved from <https://www.ren21.net>

Sustainable Agriculture Research & Education (SARE). (n.d.). Managing manure with composting. Retrieved from <https://sare.org>

The Goat Spot. (n.d.). Natural remedies for goats: Boosting immunity and reducing antibiotic dependency. Retrieved from <https://thegoatspot.net/>

United Nations Industrial Development Organization (UNIDO). (n.d.). Biogas production. Retrieved from <https://www.unido.org>

University of Arkansas Division of Agriculture. (n.d.). Rotational grazing: Benefits and strategies for small ruminants. Retrieved from <https://www.uaex.edu/>

University of California Agriculture and Natural Resources. (n.d.). Rabbit housing and care. Retrieved from <https://ucanr.edu/>

University of Florida IFAS Extension. (n.d.). Methods of composting. Retrieved from <https://edis.ifas.ufl.edu>

University of Kentucky. (2021). Feeding Practices to Minimize Feed Waste. Retrieved from <https://afs.ca.uky.edu/content/feeding-practices-minimize-feed-waste>